Euclid's Elements — Book II
Proposition 11

Formal Reconstruction — Technical Documentation

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CGJteamLab

To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.

1 Introduction

Proposition II.11 is the first construction problem of Book II whose conclusion is itself a new division point on the given segment. Starting from a nondegenerate straight line AB, Euclid constructs a point H between A and B such that the rectangle contained by the whole segment AB and the smaller part HB is equal in content to the square on the remaining part AH:

$$\mathrm{Rect}(AB, HB) = \mathrm{Sq}(AH).$$

The division is the classical extreme-and-mean-ratio construction, later reused in the construction of the regular pentagon.

The current Lean theorem has the same geometric output. It constructs an interior point H, an arbitrary rectangle representative contained by AB, HB, and a square representative on AH, and proves that the two figures are `HilbertScissorsEquicomplementable`. The theorem therefore belongs to the synthetic content layer. It does not introduce numerical area, a product of segment lengths, or a field equation.

A modern algebraic shadow is obtained by writing $AB = a$ and $AH = x$. Since $HB = a - x$, the target relation has the mnemonic form

$$a(a - x) = x^2, \qquad \text{equivalently} \qquad x^2 + ax = a^2.$$

This is explanatory notation only. Neither Euclid's proposition nor the Lean statement is an equation in a previously defined arithmetic of segment lengths.

The production source begins with

```
import CGJteamLab.HilbertRectangle
import CGJteamLab.Proposition2_2
```

```
import CGJteamLab.Proposition2_3
import CGJteamLab.Proposition2_6
import CGJteamLab.Proposition10
import CGJteamLab.Proposition16
import CGJteamLab.Proposition19
import CGJteamLab.Proposition20
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
```

Thus II.11 is the first proposition after II.10 to return directly to earlier Book II rectangle identities: the content proof explicitly calls II.2, II.3, and II.6, while the construction layer uses Book I order and metric results.

2 Euclid's Statement and Classical Configuration

2.1 The statement

In Heath's wording, Proposition II.11 states:

> To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.

With the lettering used in the current formal theorem, the required point is H with

$$A - H - B$$

and the content identity is

$$\mathrm{Rect}(AB, HB) = \mathrm{Sq}(AH).$$

The order matters. The rectangle uses the *smaller* piece HB, while the square is on the *remaining* piece AH.

Hartshorne emphasizes the same interpretation in his discussion of the regular pentagon: II.11 divides a segment so that the rectangle on the whole and the smaller part has the same content as the square on the larger part [?]. This is the content form of the extreme-and-mean-ratio condition; a ratio equation is not part of the Book II statement.

2.2 The classical construction

The figure is normalized to the vertex order used by the Lean source.

(1) On AB construct the square $ABCD$ by I.46.

(2) Bisect the side DA at E by I.10.

(3) Join E to B.

(4) Produce DA through A to a point F such that

$$EF \cong EB.$$

(5) Transfer the length AF to the original segment, obtaining H with

$$A - H - B, \qquad AH \cong AF.$$

The characteristic source order is therefore

$$D - E - A - F, \qquad A - H - B.$$

In Euclid's diagram the square on AF and the associated parallel lines are then used to identify the final square on AH and the rectangle contained by AB, HB [**?**]. The current Lean proof reaches the same point H through a different synthetic existence argument; this difference is recorded below rather than hidden inside the classical picture.

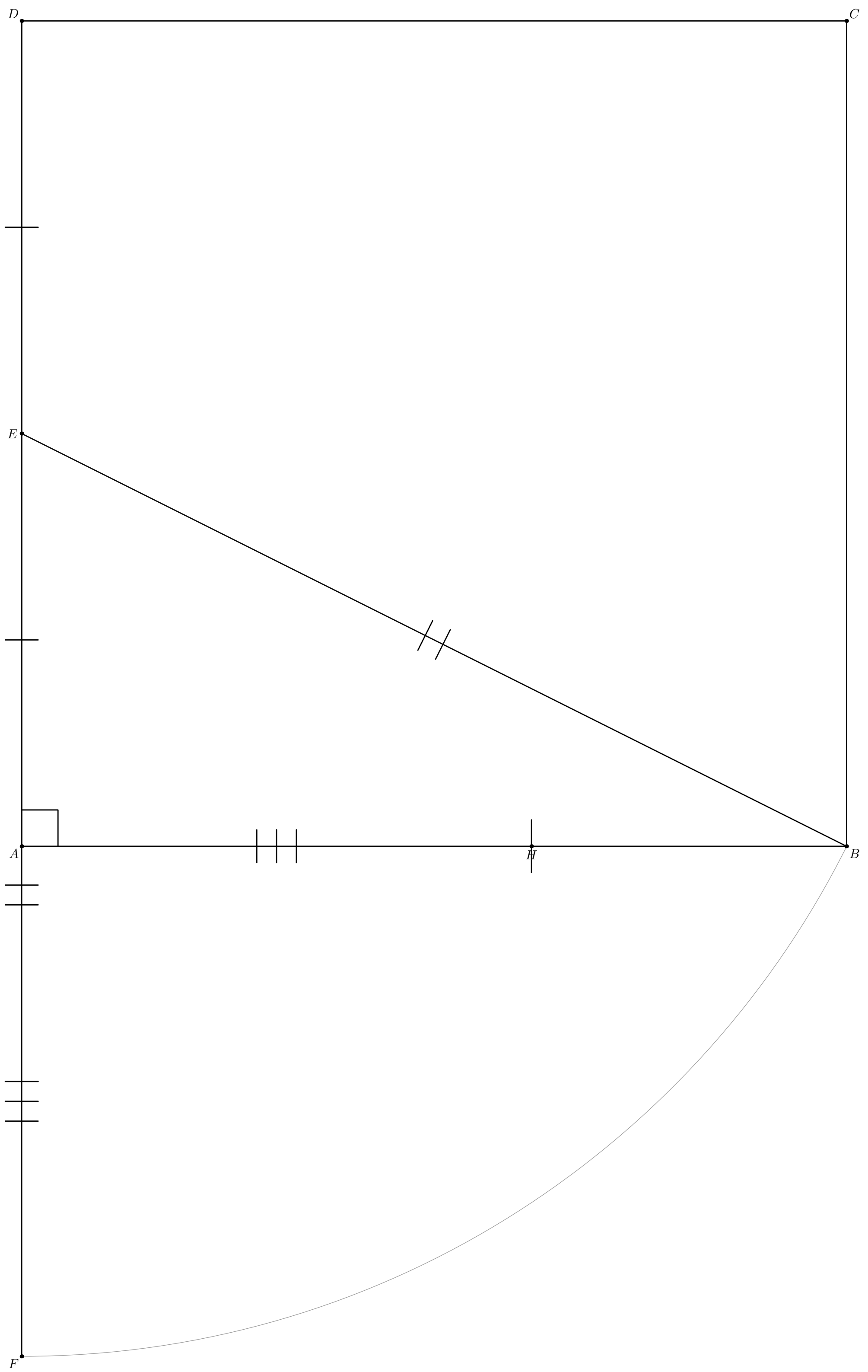

Figure 1: The principal configuration for Proposition II.11 in the lettering of the Lean source. $ABCD$ is the square on AB, E is the midpoint of DA, and F lies on the extension through A with $EF \cong EB$. The light construction arc centered at E records the classical layoff $EF = EB$. The final point H lies on AB and satisfies $AH \cong AF$. Thus the source orders are $D - E - A - F$ and $A - H - B$. The technical rectangle-cut witnesses used internally by Lean are deliberately omitted.

3 Classical Proof

The proof can be organized by comparing the exact II.6 decomposition with one Pythagorean decomposition from I.47, followed by subtraction of common figures.

Because E is the midpoint of DA and F lies beyond A, Proposition II.6 applies to the straight line DA with midpoint E and added segment AF. In the present lettering it yields

$$\mathrm{Sq}(AE) + \mathrm{Rect}(DF, AF) = \mathrm{Sq}(EF).$$

By construction $EF = EB$. The angle EAB is right because EA lies on the side AD of the square on AB. Proposition I.47 in the right triangle EAB gives

$$\mathrm{Sq}(EB) = \mathrm{Sq}(AE) + \mathrm{Sq}(AB).$$

Hence

$$\mathrm{Sq}(AE) + \mathrm{Rect}(DF, AF) = \mathrm{Sq}(AE) + \mathrm{Sq}(AB).$$

Removing the common square on AE gives

$$\mathrm{Rect}(DF, AF) = \mathrm{Sq}(AB).$$

The remaining step converts the rectangle on DF, AF into the desired rectangle on AB, HB. Euclid's square on AF and the parallel cut separate the common part geometrically. In the language of the current Book II rectangle calculus the same content bookkeeping is expressed by II.3 and II.2:

$$\mathrm{Rect}(DF, AF) = \mathrm{Rect}(DA, AF) + \mathrm{Sq}(AF),$$

while

$$\mathrm{Sq}(AB) = \mathrm{Rect}(AB, AH) + \mathrm{Rect}(AB, HB).$$

The square $ABCD$ gives $DA \cong AB$, and the construction gives $AF \cong AH$. Thus

$$\mathrm{Rect}(DA, AF) = \mathrm{Rect}(AB, AH).$$

Subtracting this common rectangle leaves

$$\mathrm{Sq}(AF) = \mathrm{Rect}(AB, HB).$$

Finally $AF \cong AH$, so the square on AF has the same content as the square on AH. Therefore

$$\mathrm{Rect}(AB, HB) = \mathrm{Sq}(AH),$$

as required.

The local classical dependency picture is therefore

$$\begin{gathered}
\text{I.46: square on } AB \quad + \quad \text{I.10: midpoint of } DA \\
\Downarrow \\
\text{I.3 / segment layoff: } EF = EB \quad + \quad \text{join } EB \\
\Downarrow \\
\text{II.6 on } D - E - A - F \quad + \quad \text{I.47 on the right triangle } EAB \\
\Downarrow \\
\text{Common Notion 3: remove the common square on } AE \\
\Downarrow \\
\text{square/rectangle decomposition on } AF \text{ and } AB \\
\Downarrow \\
\text{Common Notion 3 again} \\
\Downarrow \\
\text{II.11.}
\end{gathered}$$

4 The Geometric Identity in the Current Formalization

The public theorem takes only a nondegenerate segment:

```
(A B : Geo.Point)
(hAB : Ne A B)
```

and constructs all other points internally.

At the final interface Lean produces

$$A - H - B,$$

a concrete rectangle representative R contained by AB, HB, and a concrete square representative Q on AH. The conclusion is

$$R \simeq_{\mathrm{ec}} Q,$$

where $\simeq_{\mathrm{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This is *not* an exact `HilbertScissorsEq` conclusion. Exact dissections occur internally, in particular in the imported II.2, II.3, and II.6 blocks. The final theorem is more general because the Pythagorean comparison enters through the current I.47 equal-content interface, and the proof performs cancellation at that level.

The representation layer is also explicit. The phrase "rectangle contained by AB, HB" is represented by `IsRectangleContainedBy`; it does not identify one canonical rectangle object. Likewise the square on AH is an existentially chosen `IsSquare` representative. The theorem proves content independence from those concrete choices through the established transport machinery.

5 Reusable Book II Infrastructure Used

5.1 Proposition II.6

The first content block is the exact identity

$$\mathrm{Sq}(AE) + \mathrm{Rect}(DF, AF) \simeq_{\mathrm{sc}} \mathrm{Sq}(EF).$$

The proposition-local II.6 wrapper

```
proposition2_11_II6_exists
```

packages the large configured API of `euclid_proposition_2_6`. It constructs the required square and rectangle representatives and all explicit cut witnesses before calling II.6.

Its current signature begins:

```
private theorem proposition2_11_II6_exists
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (D A E F : Geo.Point)
    (hMidE : HilbertIsMidpoint Geo E D A)
    (hDAF : Geo.Between D A F) :
```

and returns an exact `HilbertScissorsEq`. Thus II.6 is used at the stronger dissection level before the proof enters the broader equicomplementability chain.

5.2 Propositions II.3 and II.2

The second subtraction block uses II.3 to split the rectangle contained by DF, AF at A:

$$\mathrm{Rect}(DF, AF) \simeq_{\mathrm{sc}} \mathrm{Rect}(DA, AF) + \mathrm{Sq}(AF).$$

It then uses II.2 on the square $ABCD$ cut at H:

$$\mathrm{Rect}(AB, AH) + \mathrm{Rect}(AB, HB) \simeq_{\mathrm{sc}} \mathrm{Sq}(AB).$$

Both imported propositions provide exact scissors equalities. The proposition-local proof subsequently transports the first common rectangle to the second by side congruences.

5.3 Rectangle representative transport

The square $ABCD$ yields

$$DA \cong AB,$$

while the construction of H yields

$$AF \cong AH.$$

The theorem `rectangle_transport_scissorsEq` therefore gives an exact scissors equality between representatives of

$$\mathrm{Rect}(DA, AF) \quad \text{and} \quad \mathrm{Rect}(AB, AH).$$

This is a representation theorem, not commutativity or multiplication in a segment field.

5.4 Square transport

I.47 and I.46 create independent square representatives on equal or congruent segments. The proof repeatedly uses `hilbert_square_transport` to move between them. The final use transports the square on AF to the square on AH through

$$AF \cong AH.$$

Again, this is geometric/content transport between representatives rather than an arithmetic rewrite of squared numbers.

6 New Proposition-Local Rectangle-Cut Infrastructure

A significant formal feature of II.11 is not visible in the public theorem: the current low-level APIs of II.2, II.3, and II.6 require explicit rectangular cut diagrams. Rather than make those witnesses hypotheses of the public theorem, II.11 constructs them internally.

The proposition-local chain is

```
proposition2_11_rectangle_cut_core
proposition2_11_rectangle_cut_between
proposition2_11_diagonal_cut_intersection
proposition2_11_rectangle_cut_exists
```

`proposition2_11_rectangle_cut_core` starts with an arbitrary rectangle $BCED$ and a point M satisfying $B - M - C$. It constructs an upper point L and the two parallelograms

$$LDBM, \qquad CELM.$$

The proof uses the fourth-vertex parallelogram construction and Euclidean parallel uniqueness to show that L lies on the carrier DE.

The next helper strengthens collinearity to strict order. Opposite-side congruences in the original rectangle and the two cut parallelograms give

$$BM \cong DL, \qquad BC \cong DE, \qquad MC \cong LE.$$

Hilbert's three-point order transport then sends

$$B - M - C \quad \Longrightarrow \quad D - L - E.$$

Finally `proposition2_11_diagonal_cut_intersection` uses Pasch in the nondegenerate triangle CBD. Since the cut line ML enters through CB at M and cannot meet BD because $DB \parallel ML$, it must meet CD at an interior point N. Parallel crossing order then proves

$$D - N - C, \qquad M - N - L.$$

The final local wrapper

```
proposition2_11_rectangle_cut_exists
```

packages exactly the geometric witnesses expected by the existing rectangle split APIs.

This infrastructure is private to Proposition II.11. No new theorem is added to the public `HilbertRectangle` module. The documentation therefore records it as proposition-local reusable geometry rather than as a new public rectangle API.

7 Proof Architecture of the Reconstruction

7.1 Step 1: construct the square and midpoint

Starting from $A \neq B$, the construction helper calls I.46 to produce the square $ABCD$. It then calls I.10 on the side DA and obtains a midpoint E:

$$D - E - A, \qquad DE \cong EA.$$

Opposite sides and equal sides of the square provide

$$DA \cong AB.$$

The side EA is collinear with DA, so the right angle of the square at A is transported to

$$\angle EAB$$

as a formal `HilbertRightAngle`.

7.2 Step 2: prove that the copied hypotenuse passes beyond A

Euclid's diagram makes it visually clear that the point F with $EF = EB$ lies beyond A. Lean proves the required inequality first.

The helper `proposition2_11_EA_lt_EB_aux` extends EA beyond A to a point X. I.16 in triangle BEA gives

$$\angle EBA < \angle BAX.$$

Because $E - A - X$ and EAB is right, the exterior angle BAX is congruent to the right angle EAB. Transporting the angle inequality and applying I.19 gives

$$EA < EB.$$

This is a genuine synthetic order argument. No square-root or numerical length comparison is used.

7.3 Step 3: construct F and prove the four-point order

Hilbert segment construction lays off EB on the ray EA, producing F with

$$EF \cong EB.$$

The same-ray data alone do not determine whether F lies before or beyond A. The previously proved inequality $EA < EB$, transported through $EB \cong EF$, rules out the wrong betweenness branches. Order trichotomy then gives

$$E - A - F.$$

Together with $D - E - A$ this yields

$$D - E - A - F.$$

7.4 Step 4: prove AF is shorter than AB

The formal proof does not obtain H by invoking a line-circle intersection. Instead it proves

$$AF < AB.$$

The helper `proposition2_11_AF_lt_AB_aux` applies I.20 in triangle AEB. It obtains a point T with

$$E - A - T, \qquad AT \cong AB, \qquad EB < ET.$$

Using $EF \cong EB$, the inequality becomes $EF < ET$. Unpacking the definition of strict segment comparison gives a point P on ET with $EP \cong EF$. Uniqueness of segment construction on the same ray identifies P with F. Hence

$$A - F - T,$$

so $AF < AT$. Transport through $AT \cong AB$ yields

$$AF < AB.$$

7.5 Step 5: obtain the final cut H

By definition, `HilbertSegmentLessGeoAFAB` contains a point of the segment AB at distance AF from A. The helper `proposition2_11_construct_H` unpacks that witness and returns

$$A - H - B, \qquad AF \cong AH.$$

This is the main construction-level divergence from the classical source. Hartshorne notes that a line-circle intersection property is used in Euclid's II.11 construction [?]. The Lean proof reaches the same metric cut through segment comparison and segment-construction uniqueness. It is therefore synthetic and source-compatible in result, but not a line-by-line formalization of the historical construction.

7.6 Step 6: the first II.6–I.47 comparison

The helper `proposition2_11_central_equality` combines two content paths.

First, II.6 gives the exact scissors identity

$$S_{AE} + R_{DF,AF} \simeq_{\mathrm{sc}} S_{EF}.$$

Second, I.47 in the right triangle AEB gives

$$S_{EB} \simeq_{\mathrm{ec}} S_{AE} + S_{AB}.$$

Since $EF \cong EB$, square transport replaces S_{EB} by S_{EF}. Hence

$$S_{AE} + R_{DF,AF} \simeq_{\mathrm{ec}} S_{AE} + S_{AB}.$$

7.7 Step 7: Common Notion 3 in the scissors calculus

The local helper

```
private theorem proposition2_11_common_notion_3
    (P Q R : HilbertScissorsTerm Geo)
    (hWhole :
      HilbertScissorsEquicomplementable Geo
        (R + P)
        (R + Q)) :
    HilbertScissorsEquicomplementable Geo P Q := by
```

formalizes the content-level cancellation needed by Euclid.

It does not assert subtraction in a numerical group. It unfolds the witness structure of `HilbertScissorsEquicomplementable`: if $R + P$ and $R + Q$ become scissors-equal after adding complements U, V, then P and Q become scissors-equal after adding the enlarged complements $R + U, R + V$. Associativity and commutativity of the finite formal sum provide the bookkeeping.

Applied to the previous comparison, this removes the common square on AE and gives

$$R_{DF,AF} \simeq_{\mathrm{ec}} S_{AB}.$$

7.8 Step 8: the second subtraction

The helper `proposition2_11_second_subtraction` now combines three exact-scissors statements:

$$\begin{aligned} R_{DF,AF} &\simeq_{\mathrm{sc}} R_{DA,AF} + S_{AF} && \text{by II.3,} \\ R_{AB,AH} + R_{AB,HB} &\simeq_{\mathrm{sc}} S_{AB} && \text{by II.2,} \\ R_{DA,AF} &\simeq_{\mathrm{sc}} R_{AB,AH} && \text{by rectangle transport.} \end{aligned}$$

Together with the central equicomplementability

$$R_{DF,AF} \simeq_{\mathrm{ec}} S_{AB},$$

these yield

$$R_{DA,AF} + S_{AF} \simeq_{\mathrm{ec}} R_{DA,AF} + R_{AB,HB}.$$

The second Common-Notion-3 helper cancels the common rectangle:

$$S_{AF} \simeq_{\mathrm{ec}} R_{AB,HB}.$$

7.9 Step 9: final square transport

I.46 constructs the target square on AH. Since $AF \cong AH$, `hilbert_square_transport` gives

$$S_{AF} \simeq_{\mathrm{ec}} S_{AH}.$$

Reversing the relation from Step 8 and composing by transitivity gives

$$R_{AB,HB} \simeq_{\mathrm{ec}} S_{AH}.$$

This is exactly the public II.11 conclusion.

8 Lean Formalization

8.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_11
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B : Geo.Point)
    (hAB : Ne A B) :
    exists
      H
      R0 R1 R2 R3
      Q0 Q1 : Geo.Point,
      Geo.Between A H B /\
      IsRectangleContainedBy Geo
        R0 R1 R2 R3 A B H B /\
      IsSquare Geo A H Q0 Q1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo R0 R1 R2 R3)
        (hilbertParallelogramTerm Geo A H Q0 Q1) := by
```

The theorem is deliberately existential at the representation level. It does not expose the auxiliary square $ABCD$, the midpoint E, the extension point F, the intermediate squares used by I.47, or the cut witnesses required by II.2, II.3, and II.6.

8.2 The proposition-local theorem chain

The production module is factored into

```
proposition2_11_rectangle_cut_core
proposition2_11_rectangle_cut_between
proposition2_11_diagonal_cut_intersection
proposition2_11_rectangle_cut_exists
proposition2_11_II6_exists
proposition2_11_common_notion_3
proposition2_11_central_equality
proposition2_11_common_notion_3_second
proposition2_11_second_subtraction
proposition2_11_final_transport
proposition2_11_EA_lt_EB_aux
proposition2_11_construct_F_aux
proposition2_11_AF_lt_AB_aux
proposition2_11_construct_H
euclid_proposition_2_11
```

The source order places the content infrastructure before the final construction helpers, but the mathematical dependency graph is more naturally read in the construction-first order used in the preceding section.

8.3 The public wrapper

The public theorem first calls `proposition2_11_construct_H`, obtaining

$$C, D, E, F, H$$

together with the square, midpoint, the orders $D - E - A$ and $E - A - F$, $EF \cong EB$, $AF < AB$, $A - H - B$, $AF \cong AH$, noncollinearity, and the right angle EAB.

It then combines $D - E - A$ and $E - A - F$ to derive $D - A - F$ and passes the complete geometric package to `proposition2_11_final_transport`. Only the point H, the target rectangle representative, the target square representative, and the final equicomplementability witness escape the wrapper.

9 Relationship with Earlier Book II Propositions

Proposition II.11 directly reuses three earlier Book II theorems:

$$\text{II.2}, \qquad \text{II.3}, \qquad \text{II.6}.$$

II.6 supplies the first major normal form:

$$S_{AE} + R_{DF,AF} \simeq_{\text{sc}} S_{EF}.$$

II.3 decomposes the rectangle $R_{DF,AF}$ at A, and II.2 decomposes the square S_{AB} at the final cut H. Thus II.11 is not merely a Book I construction followed by Pythagoras. Its final content argument genuinely depends on the rectangle calculus developed earlier in Book II.

The role of II.6 is especially source-faithful: Euclid himself invokes II.6 in the classical proof of II.11. The uses of II.2 and II.3 are more architectural. They provide current formal normal forms for the final subtraction that Euclid performs directly in his configured square-and-rectangle diagram.

There is no direct theorem-level dependency on II.9 or II.10. II.11 therefore forms a new branch: it combines the earlier exact rectangle calculus of II.1–II.8 with the I.47 equicomplementability layer that became prominent in II.9–II.10.

10 Relationship with Book I Infrastructure

The construction and proof use a broad but classical Book I layer.

- I.46 constructs the initial square on AB and later the target square on AH.
- I.10 supplies the midpoint E of DA.
- I.16 and I.19 prove the strict metric comparison $EA < EB$ needed to orient the point F.
- I.20 proves the comparison leading to $AF < AB$.
- I.47 supplies the Pythagorean content identity in triangle AEB.
- standard congruence, right-angle, same-ray, betweenness, and parallelogram facts transport these statements between the concrete representatives.

The generic rectangle-cut helper additionally uses Euclidean parallel uniqueness, Pasch, parallelogram opposite-side congruences, and a parallel crossing-order theorem. These are library-level geometric facts, not new axioms introduced by II.11.

11 Formalization Notes and Hidden Geometric Data

11.1 The source order D-E-A-F is proved

The printed construction invites the reader to see immediately that the point F lies beyond A. Lean must prove this. It first establishes $EA < EB$, constructs F with $EF = EB$, and then uses same-ray information and betweenness trichotomy to isolate the branch $E - A - F$.

Consequently the four-point order

$$D - E - A - F$$

is a theorem of the construction, not an assumption copied from a diagram.

11.2 The final point H is not obtained by a diagrammatic intersection

The source construction naturally transfers AF onto AB. Hartshorne notes that the line-circle intersection property is relevant to Euclid's II.11 construction [?]. The present formal proof instead proves $AF < AB$ and then obtains H by unpacking the strict segment comparison.

This avoids importing an additional line-circle construction into the public proof while preserving the intended synthetic geometry.

11.3 Noncollinearity carries the right-angle information

The proof of I.47 requires the right triangle AEB to be formally nondegenerate. The point E lies on the side DA of the square, and the square already gives that D, A, B are noncollinear. Collinearity transport therefore yields

$$\neg\,\mathrm{Collinear}(E, A, B).$$

The right angle at A is then transported from the square arm AD to the same ray AE.

11.4 Rectangle cuts carry hidden witnesses L and N

The imported Book II APIs do not accept only the statement that a rectangle is cut at a point. They require a complete configured diagram: an upper cut point, a diagonal intersection, two strict betweenness relations, and two parallelograms.

The private rectangle-cut constructor creates these witnesses by parallel geometry and Pasch. They are mathematically meaningful because they certify that the formal scissors split corresponds to an actual geometric cut, but they do not belong in the public II.11 statement.

11.5 Common Notion 3 is not numerical subtraction

II.11 uses content cancellation twice. This is the most important semantic difference from II.10, whose final bridge was forward-only composition.

The local Common-Notion-3 theorem does not introduce additive inverses for scissors terms. It enlarges the existing equicomplementability witnesses so that the common figure is absorbed into the complements. The operation is therefore legitimate inside the positive finite-content calculus and should not be described as subtraction of real-valued areas.

11.6 Square and rectangle representatives are noncanonical

The construction, II.6, I.47, II.2, and II.3 may create different concrete figures representing the same side data. The formal proof must explicitly move among these representatives with square and rectangle transport. This is why the public theorem existentially quantifies the final figures rather than pretending that "the rectangle" or "the square" is a canonical object.

12 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. The full project build was run after integrating II.11 and completed successfully with 1502 jobs.

The transitive axiom audit

```
#print axioms Geometry.euclid_proposition_2_11
```

reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise statement is therefore: II.11 adds no new axiom and no `sorry`; it is not literally axiom-free.

The semantic layers are

$$
\begin{gathered}
\text{incidence/order/congruence and Euclidean parallel geometry} \\
\Downarrow \\
\text{square, midpoint, segment construction, strict segment comparison} \\
\Downarrow \\
\text{concrete rectangle and square representatives} \\
\Downarrow \\
\text{exact II.6 / II.3 / II.2 scissors identities} \\
\Downarrow \\
\text{I.47 equicomplementability + representative transport} \\
\Downarrow \\
\text{two Common-Notion-3 cancellations} \\
\Downarrow \\
\text{II.11 equicomplementability conclusion.}
\end{gathered}
$$

No numerical area function is used. No segment multiplication is used. The modern golden-ratio equation is an interpretive shadow only.

13 Dependency Summary

The actual production chain is

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\text{I.46 square} + \text{I.10 midpoint} \\
\Downarrow \\
\text{I.16} + \text{I.19: } EA < EB \\
\Downarrow \\
\text{segment construction: } EF = EB, \quad E - A - F \\
\Downarrow \\
\text{I.20} + \text{segment-construction uniqueness: } AF < AB \\
\Downarrow \\
\text{construct } H:\ A - H - B,\ AF = AH \\
\Downarrow \\
\text{local rectangle-cut constructor for configured Book II APIs} \\
\Downarrow \\
\text{II.6} + \text{I.47} + \text{square transport} \\
\Downarrow \\
\text{Common Notion 3: } R_{DF,AF} \simeq_{\text{ec}} S_{AB} \\
\Downarrow \\
\text{II.3} + \text{II.2} + \text{rectangle transport} \\
\Downarrow \\
\text{Common Notion 3: } S_{AF} \simeq_{\text{ec}} R_{AB,HB} \\
\Downarrow \\
\text{square transport } AF = AH \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_11}.
\end{gathered}
$$

The implementation checkpoint is:

New geometric axiom in II.11:	no
New content axiom in II.11:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.2, II.3, II.6
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is equal-content / equicomplementability:	yes
Uses exact scissors internally:	yes
Uses content cancellation:	yes, twice
Uses numerical area:	no
Uses segment multiplication:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean, 1502 jobs
Proposition-local `sorry`/`admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

14 Conclusion

Proposition II.11 combines the two main proof modes developed so far in Book II. Its geometric construction returns to the classical square, midpoint, produced segment, and right-triangle machinery of Book I. Its content proof then reuses the exact rectangle calculus of II.2, II.3, and II.6 together with the I.47 equicomplementability and square-transport machinery that became central in II.9 and II.10.

The most important new formal feature is cancellation. Twice, Euclid removes a common figure. Lean realizes each instance directly inside the equicomplementability definition; no additive inverse, numerical area, or segment field is introduced.

The second important feature is constructional. Euclid's diagram locates the cut point by the classical extension/layoff mechanism. The current Lean proof makes the required order explicit: it proves $EA < EB$, then $AF < AB$, and only then obtains H with $A - H - B$ and $AH = AF$. This is not a line-by-line replay of the source construction, but it is a fully synthetic reconstruction of the same geometric result.

The final theorem therefore states exactly the appropriate Book II content claim:

$$\mathrm{Rect}(AB, HB) \simeq_{\mathrm{ec}} \mathrm{Sq}(AH),$$

with no strengthening to exact scissors equality and no reduction to a numerical golden-ratio equation.